

Influence of the charges deposition on the spatio-temporal self-organization of streamers in a DBD

S Celestin^{1,2}, G Canes-Boussard¹, O Guaitella¹, A Bourdon² and A Rousseau¹

¹ Ecole Polytechnique LPTP, CNRS UMR 7648, 91 128 Palaiseau, France

² Ecole Centrale Paris EM2C, CNRS UPR 288, 92 295 Châtenay-Malabry, France

Received 10 July 2008, in final form 1 September 2008

Published 1 October 2008

Online at stacks.iop.org/JPhysD/41/205214

Abstract

At atmospheric pressure, dielectric barrier discharges (DBDs) are mainly constituted of unstably triggered non-equilibrium transient plasma filaments, also called microdischarges. A self-organization of plasma filaments and their interaction with the deposited charges upon the dielectric material has been studied previously for frequencies on the order of a few kilohertz (for example see Guikema *et al* (2000 *Phys. Rev. Lett.* **85** 3817) and Chirokov *et al* (2004 *Plasma Sources Sci. Technol.* **13** 623–35)). In this paper we realized a DBD in the configuration cylinder-tip to dielectric plate biased by a low frequency power supply (50 Hz) in synthetic air. Using a CCD camera coupled with electrical diagnostics we observed collective behaviour of plasma filaments in the gas gap, as already observed for asymmetric surface dielectric barrier discharges (ASDBD) in Allegraud *et al* (2007 *J. Phys. D: Appl. Phys.* **40** 7698–706) and predicted in Guaitella *et al* (2006 *J. Phys. D: Appl. Phys.* **39** 2964–72). We also show how the surface charge deposited upon the dielectric plate modifies the spatial organization of microdischarges in this experimental device. In parallel, we made an electrostatic model of the experimental device which gave results consistent with the experimental measurements and with the assumption we made on the role of the deposited charges for their impact on the subsequent discharges during a positive half-cycle.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Dielectric barrier discharges (DBDs), also known as barrier discharges (BDs) or silent discharges, have been studied since the invention of the ozonizer by Siemens in 1857. Nowadays DBDs are widely used for industrial applications [1, 2], such as gas depollution, surface treatment, excimer lamps, ozone generation and flat plasma displays. At atmospheric pressure (for gaseous gaps on the order of a few millimetres), DBDs are mainly constituted of unstably triggered non-equilibrium transient plasma filaments (also called microdischarges) [2]. The short duration (a few tens of nanoseconds) and unpredictable triggering of these filaments make them difficult to study experimentally. During the last decade, a few authors have succeeded in following the propagation of a single filament (the streamer phase) with high resolution in time and space in a double

dielectric barrier configuration, and thus have determined the electron density and electric field inside using spectroscopic diagnostics [3].

In [4] Guikema *et al* observed self-organized patterns in a one-dimensional symmetric DBD system with He/Ar mixtures and associated the patterns with characteristics for charge transfer during one cycle of the applied voltage (for frequencies on the order of a few kilohertz). They concluded that the patterns resulted from the presence of surface charge deposits; that is, related to the intrinsic so-called memory effect of DBDs (e.g. see [2, 5]). Klein *et al* [6] progressed further in this study by improving the time resolution of the imaging technique and observed that the patterns obtained in [4] were in fact constituted of several filaments ignited at different times, which were spatially and temporally controlled by the residual surface charges on the dielectric. It is also interesting to note that Chirokov *et al* also demonstrated in [7] behaviour of

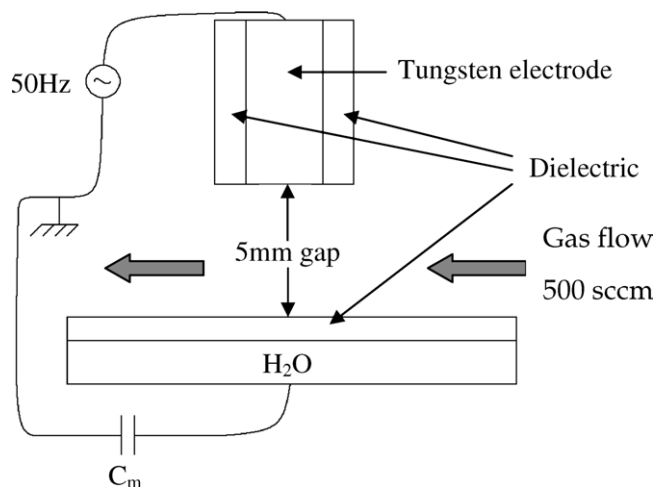


Figure 1. Scheme of the reactor.

self-organization in a plane-to-plane DBD in air at atmospheric pressure driven by high-frequency sinusoidal voltage (kHz) through a comparison of experimental results with a numerical model.

Recently, Guaitella *et al* [8] described the bimodal behaviour of the statistical distribution of current peaks in a cylindrical DBD, and they concluded that the high-current group was due to the self-triggering of several filaments, presumably correlated by a radiative effect called the ‘collective effect’. They also studied the impact of this collective effect on the injected energy in the discharge, which directly controls the chemistry. Collective behaviour of microdischarges has also been recently observed by imaging with a CCD camera on asymmetric surface dielectric barrier discharges (ASDBD) in [9].

This paper seeks to demonstrate such a collective effect along with the presence of patterns of microdischarges for a volume DBD in air at atmospheric pressure at low frequency (50 Hz) using a new experimental device coupled with an imaging method.

Firstly, we describe the experimental setup and the diagnostics in section 2. In section 3 we show the characteristic regimes of the DBD by a statistical study. To show a clear correlation between discharges corresponding to different peaks and the collective behaviour of the microdischarges we present CCD imaging results of individual current peaks during the positive half-cycle in section 4. Then we support the argument that the deposited charges on the dielectric plate are responsible for the spatial and temporal organization of the discharges in sections 5 and 6.

2. Experimental setup

2.1. DBD reactor

The discharges are produced in a Pyrex cell (figure 1). The upper electrode is a cylinder in tungsten (diameter 2 mm) coated with a 1 mm layer of dielectric except on its tip, and it is linked to a 50 Hz sine wave high-voltage power supply. The lower electrode is salt water to allow for imaging. The

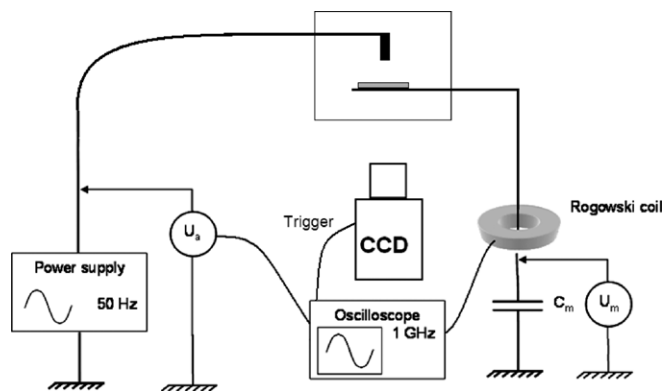


Figure 2. Experimental set-up scheme.

discharge was not influenced by the salt concentration or the resulting change in the resistance of the water. In the following the conductivity of the water is $\sim 800 \mu\text{S cm}^{-1}$.

The dielectric is a cylindrical Pyrex plate of 2.2 cm diameter and 2 mm thickness (figures 1 and 2) and placed above the water. Dry synthetic air flows at 500 sccm (standard cubic centimetre) through the 5 mm gap at atmospheric pressure. A gauge controls the pressure downstream of the reactor.

2.2. Electrical diagnostic

At atmospheric pressure in air, breakdown occurs at about 13 kV in this reactor. This study was performed for three applied voltage amplitudes V_a of 15, 18 and 20 kV.

CCD imaging is performed by an Andor iStar 734 camera, via a B7838-UV Pentax objective which is efficient in the spectral range 230–800 nm. The Lecroy LT584M 1 GHz 4Gs/s oscilloscope, which can output an electric signal upon detecting a rising front, is used to trigger the camera. The camera can be positioned either ‘underneath’ the reactor (as sketched in figure 2) to image the plasma zone through the water or beside the reactor to image the gap. Current peaks are measured with a Fischer F33-5 Rogowski coil with a bandwidth from 1 to 100 MHz. A capacitor $C_m = 1.11 \text{ nF}$ is connected in series with the reactor to measure injected energy and charge transfer. Both V_a and V_m are measured with Lecroy PPE20kV probes (figure 2). The DBD generates a small number of very short current peaks, measured as voltage steps across the capacitor as shown in figure 3, due to the ignition of microdischarges between the electrodes that are extinguished by the screening from the charge deposited on the dielectric plate (this process occurs over a few tens of nanoseconds) [1, 2]. The first and second current peaks correspond to the first and second voltage steps, respectively. Although DBDs generally exhibit a great number of current peaks, in this configuration only a few current peaks are observed (up to 4).

Note that in such an experimental configuration, the electrical signal of the capacitor voltage possesses a jitter of about 1 ms because of the usual unpredictability of AC DBDs at atmospheric pressure and low frequency without specific preparation of the high-field electrode [10]. Moreover, one or several current peaks occur during the positive half-cycle.

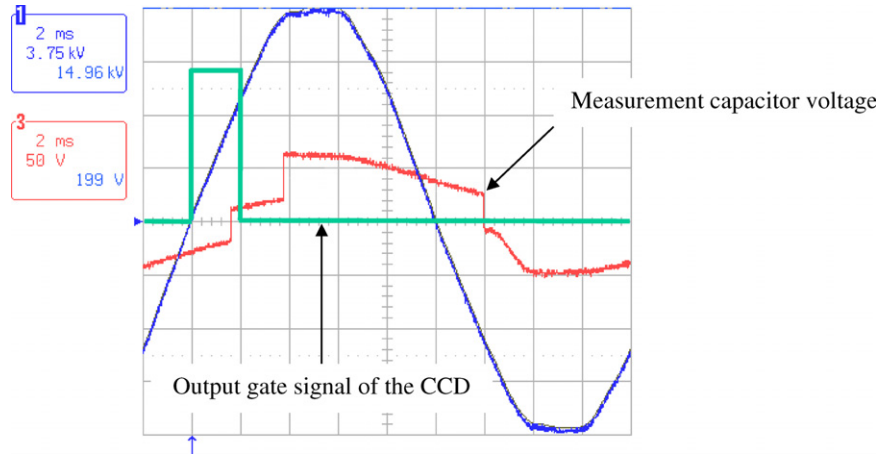


Figure 3. Screenshot of the oscilloscope. Sinusoidal curve: power supply voltage, 3.75 kV div^{-1} . Measurement capacitor voltage: 50 V div^{-1} . Output gate signal of the CCD: arbitrary units. Timebase: 2 ms div^{-1} .

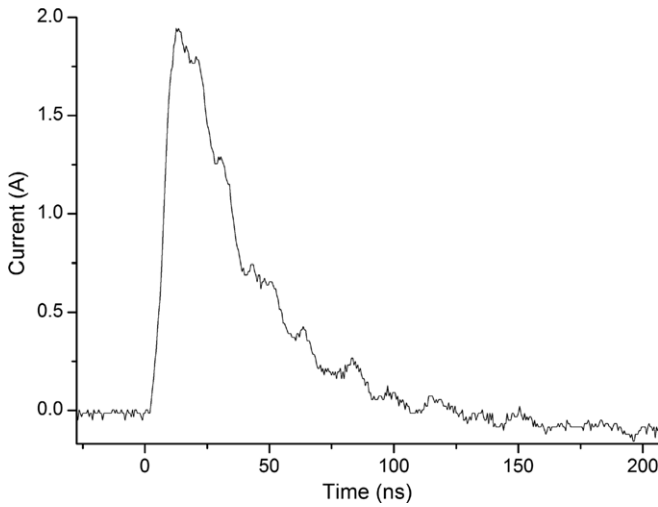


Figure 4. Current peak measured by the Rogowski coil.

The total number of current peaks in one positive half-cycle depends on the applied voltage.

Figure 4 shows an example of a characteristic current peak as a single-shot record, corresponding to the first step on the measurement capacitor voltage, measured with the Rogowski coil. We can see that its total duration is about 100 ns and its maximum is about 2 A. After $\sim 100 \text{ ns}$ the current becomes slightly negative ($\sim -0.1 \text{ A}$), we assumed that this was due to perturbations in the electrical circuit.

Figure 3 also shows the output gate signal of the CCD camera. The CCD array is opened when the signal is non-zero. When the power supply changes polarity, the CCD switches on for 2 ms in order to overlap with the first positive step of the measured capacitor voltage. The aim of this work is to detect an event corresponding to an individual current peak (one step in the capacitor voltage) and to record the current peak and the corresponding image. Given that few current peaks occur during the positive half-cycle, a CCD gate of 2 ms results in the detection of only a single current peak. Sometimes, two current peaks can appear during the opening of the CCD, but such images are then discarded.

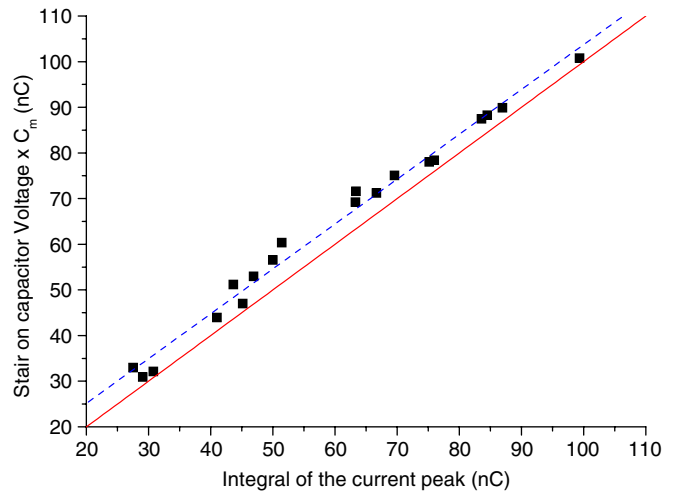


Figure 5. Closed squares: measured integration of current peak versus the capacitor charge. Dashed line: the linear regression. Solid line: representation of the function $y = x$.

2.3. Current measurement and charge transfer

The transferred charge is estimated in two ways:

- The voltage increase at the measurement capacitor, using $Q = C_m \cdot \Delta U_m$,
- The integration of the current peak, using $Q = \int i(t) \cdot dt$ from $t = 0$ to $t(i = 0)$, for a current peak recorded in single-shot.

Figure 5 shows that the two measurement methods give close results.

We can see that the charge measured using the integral of the current peak is generally less than the charge measured with the capacitor. This is due to the sampling of the current peak (recorded in single-shot) by the oscilloscope; the integral of the current peak (b) is then slightly underestimated.

2.4. Injected power

We measured the mean power injected in the gas by the discharge for a given applied voltage amplitude using the

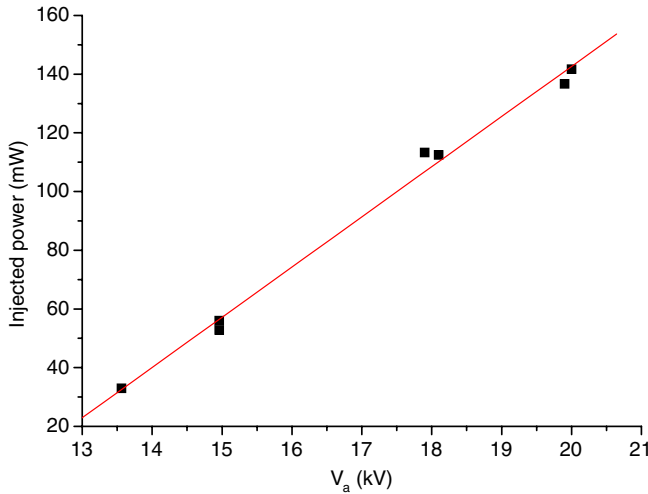


Figure 6. Mean injected power measured using the Manley method versus the applied voltage amplitude V_a .

Manley method [11] (or Lissajous method) as described in [9] and [8].

Figure 6 shows that the mean injected power is a linear function of the maximum voltage. The injected power increases by about 17 mW per kilovolt. Compared with the results obtained in [8], the mean injected power we obtained here is less by a factor of ten. This is related to the difference in the reactor geometries used; for example, it is interesting to note that the zone where the plasma takes place in the reactor used in [8] is about ten times larger than the one used for this study.

3. Current peak statistics

3.1. Temporal occurrence of current peaks

In order to characterize the DBD, a statistical study over ignition times of current peaks was carried out for different applied voltage amplitudes V_a . Figures 7(a) and (b) present the mean number of peaks averaged over 100 periods of applied voltage, during each millisecond increment of the period, starting from $t = 0$ up to 20 ms (the period of the power supply voltage), for applied voltage amplitudes of 15 and 18 kV; $t = 0$ refers to the voltage zero value, increasing voltage.

Figure 7(a) shows that at $t = 2.5 \pm 0.5$ ms, there is a 96% chance of measuring a current peak and the corresponding discharge. Figure 7(a) also shows the fluctuation in the moment of appearance of the current peaks. We notice two time intervals for their appearance in the positive half-cycle ($t = 0$ –10 ms). Current peaks appear more consistently in time during the positive half-cycle in figure 7(b), which shows the measurements for the case of $V_a = 18$ kV.

But figures 7(a) and (b) cannot detail the probability of the total number of current peaks in the overall half-cycle nor the influence of the applied voltage on this probability.

3.2. Number of current peaks per half-cycle

The relative number of current peaks occurring in the positive half-cycle is shown in figure 8 for power supply voltage

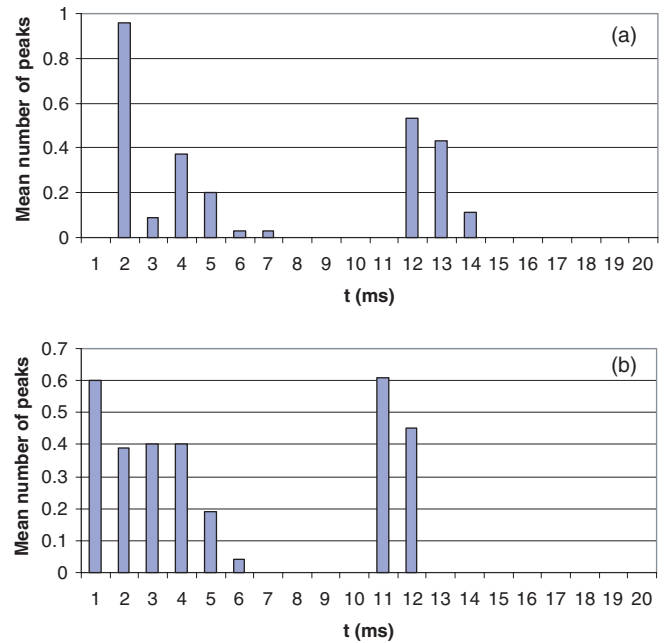


Figure 7. Mean number of current peaks per millisecond from $t = 0$ ms to $t = 20$ ms. Between 10 and 20 ms, the current peaks are negative. (a) $V_a = 15$ kV and (b) $V_a = 18$ kV.

amplitudes of 15, 18 and 20 kV. Figure 8 was obtained by counting the total number of events (1, 2, 3 or 4 current peaks during the positive half-cycle) over 100 complete positive half-cycles of the applied voltage, unlike the incremental method used to obtain figures 7(a) and (b).

For 15 kV, figure 8 shows that two situations are observed in this regime: either only one or two current peaks occur during the positive half-cycle, with the latter in the majority of cases. For the two current peaks case, there is about 2 ms between peaks (see figure 3). Similar results for 18 and 20 kV are presented in figure 8.

The maximum of occurrences obtained for every V_a is two current peaks; the cases of 3 and 4 total current peaks only occur when the amplitude is 18 or 20 kV. There was only one current peak during the negative half-cycle for all the studied applied voltages.

From the results presented in figure 8, we can estimate an average number of current peaks during the positive half-cycle for each applied voltage amplitude (see table 1). The data are consistent although insufficient to evaluate a real tendency. Indeed, as expected for a DBD, the number of events increases with the applied voltage amplitude.

One can estimate a ‘statistical error’ by summing the probabilities obtained in figures 7(a) and (b) and dividing by the average total number of peaks. For 15 and 18 kV the error bars are shown in figure 8. These error bars are small (less than 10%) and show the consistency of the statistical approach used in figures 7 and 8.

3.3. Influence of first event on subsequent events

The measured current amplitude of first current peaks during the positive half-cycle when V_a is 18 kV is plotted as a function

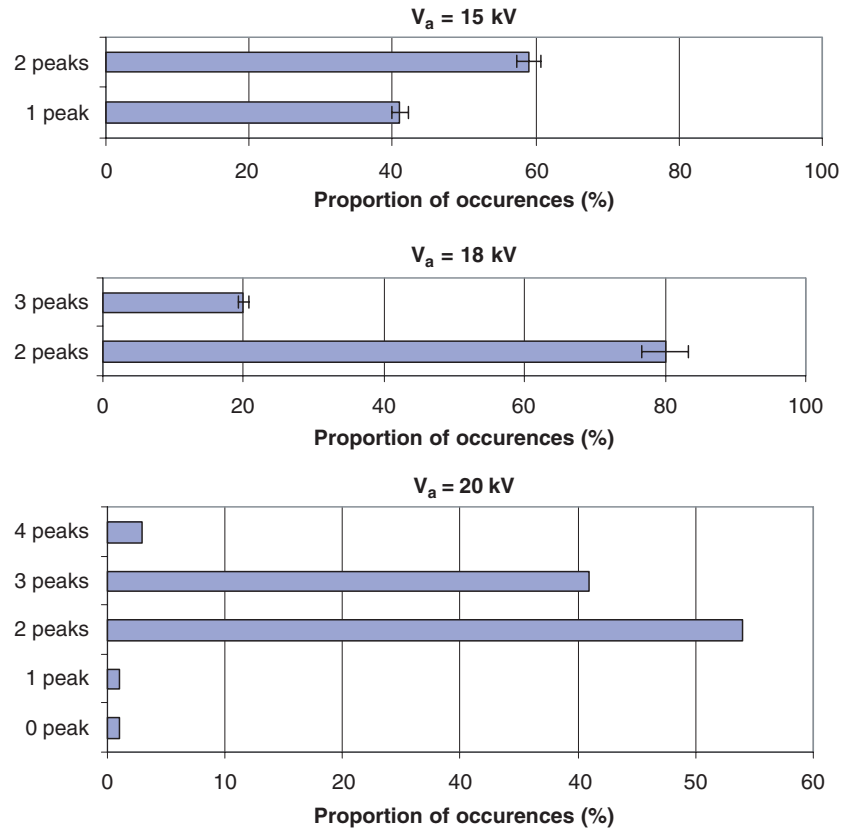


Figure 8. Number of observed total current peaks by percentage during the positive half-cycle for power supply voltage amplitudes of 15, 18 and 20 kV.

Table 1. Average total number of current peaks during the positive half-cycle.

V_a (kV)	15	18	20
$\langle N \rangle$	1.6	2.2	2.4

of V_{peak} , the ignition voltage of a first current peak, in figure 9. The accuracy of the maximum of a current peak depends on the sampling rate of the current peak by the oscilloscope. Error bars are then evaluated according to the sampling rate. Higher sampling rates result in smaller error, but lower sampling rates permit a larger time range.

One can clearly see from figure 9 that the maximum of first current peak increases with ignition voltage, and the same trend is observed for the transferred charge corresponding to the first current peaks. This is not clearly observed for the subsequent peaks. This is explained further in section 6.

To exhibit the influence of the first current peak on the subsequent ones for each positive half period we plot in figure 10 the number of peaks per positive half period as a function of the ignition voltage of the first one.

Figure 10 shows that the subsequent peaks (second, third and fourth) are strongly influenced by the first. On average, if the first current peak is ignited at small applied voltages, the subsequent events can be numerous; it becomes possible to have two, three or four peaks in total. Furthermore, according to figure 9, the total number of current peaks is inversely related to the intensity of the first. Figures 9 and 10 show that the higher the ignition voltage of the first peak, the higher

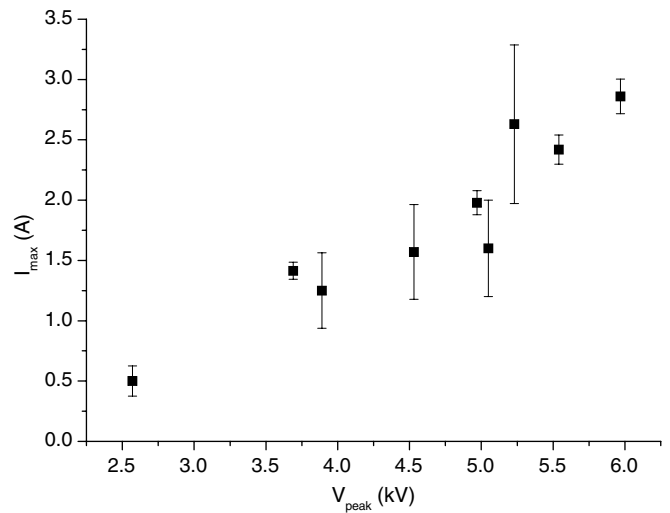


Figure 9. Measured current amplitude of first current peaks as a function of the applied voltage at which they occur (V_{peak}). Applied voltage amplitude: $V_a = 18$ kV.

its corresponding maximum current and charges transferred and the smaller the number of following peaks during this positive half-cycle. This result is explained further in section 6.

The spatial behaviour of the microdischarges is described in the following section using CCD images corresponding to individual current peaks (first, second, etc).

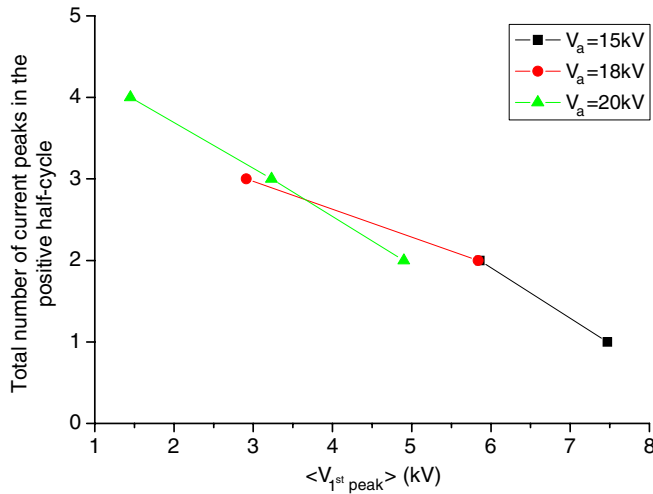


Figure 10. Total number of current peaks in the positive half-cycle for three applied voltage amplitudes as a function of the mean ignition voltage of the first current peak.

4. CCD imaging of individual current peaks during the positive half-cycle

4.1. Methodology

In the previous section we showed that, depending on the ac voltage, a statistical number of peaks appears during the positive half-cycle. Here, we do not monitor the propagation of the plasma filaments. CCD imaging has been used to take pictures of individual current peaks (first, second, etc) to study the influence of their ranks (i.e. of their order of appearance in the positive half-cycle).

The total transmission time of the trigger signal from the oscilloscope to the CCD was estimated accurately to be 200 ns.

A CCD gate of 1 ms captures microdischarge emission corresponding to an individual current peak. The CCD gate is synchronized to a reference of zero applied voltage by triggering the CCD with the oscilloscope and can be shifted in time by introducing a delay, in order to picture the current peaks of different ranks.

Events pictured can be preceded or followed by other events. They always correspond to an individual current peak and are not accumulated.

Triggering the CCD with the rising current peak, as opposed to the zero voltage reference, always results in a blank picture for gate times ranging from 10 ns up to the time at which a subsequent current peak is measured. Thus no luminous events occur 200 ns after the rising current peak trigger time, thereby eliminating the possibility that microdischarges go undetected by the current probe.

4.2. Imaging results

As mentioned previously, only a few current peaks occur during the positive half-cycle (up to four for the 20 kV amplitude case). In the comparisons of plasma filament behaviour that follow, it will be seen that the order of occurrence (i.e. rank) of the current peak is very important. The

colour scales of figures 11 and 12 are represented in arbitrary units (the red being more intense than the violet).

4.2.1. Images of a ‘first current peak’. Figure 11 shows the luminous event corresponding to a first current peak in the positive half-cycle when $V_a = 18\text{ kV}$. In this case, three current peaks can take place in total during the positive half-cycle (two discharges follow the presented one). The picture on the left of figure 11 corresponds to a current peak of 0.35 A, which involves a charge transfer of 10 nC; the picture on the right corresponds to a current peak of 0.9 A, which involves a charge transfer of 45 nC.

Figure 11 is representative of discharges corresponding to a first current peak. They are localized around the centre of the dielectric for the underneath visualization and most of the time correspond to one or two filaments close to the central axis. We can also see on the left image of figure 11 that thin luminous filaments emanate on the dielectric surface from the ‘footprint’ of the filament crossing the gas gap (see also [12]). The figure obtained is close to the typical Lichtenberg figure for a positive charge deposition ([13] and references therein, [14, 15]). The area of this Lichtenberg figure is considerably larger than the luminous channel diameter as mentioned in [16].

4.2.2. Images of a ‘second current peak’. Images of a second current peak in the positive half-cycle were obtained in the same manner as for the first current peak. Figure 12 shows the features of such an event, for $V_a = 18\text{ kV}$ and for three current peaks in total in the positive half-cycle. The picture on the left of figure 12 corresponds to a current peak of 2.25 A, which involves a charge transfer of 90 nC; the picture on the right corresponds to a current peak of 1.5 A, which involves a charge transfer of 76 nC.

One can see that the thick filaments from the left-hand image of figure 12 correspond to the filaments in the gaseous gap from the right-hand image. The ends of the thick filaments on the left-hand image are in fact the ‘footprints’ of filaments. Hence, one can define the ‘mean impact radius’ from the centre of the dielectric to the filaments footprints as sketched in figure 12. If the picture contrast is increased, one sees that yet more filaments emanate on the surface from the footprints up to the dielectric edge, forming a Lichtenberg figure for each filament in the volume.

For each V_a studied, figure 12 is representative of microdischarges corresponding to a second current peak. The third and fourth current peaks also have this typical multi-filament and organized structure, which is quite different from the first current peaks. This behaviour of microdischarges is discussed in section 6.

4.3. Spatial and temporal self-organization—relation between filament length and ignition voltage of a current peak

It is possible to analyse the relation between the layout of the filaments and the electrical measurements by comparing the CCD camera pictures and the current peaks measured by the capacitor C_m or the Rogowski coil.

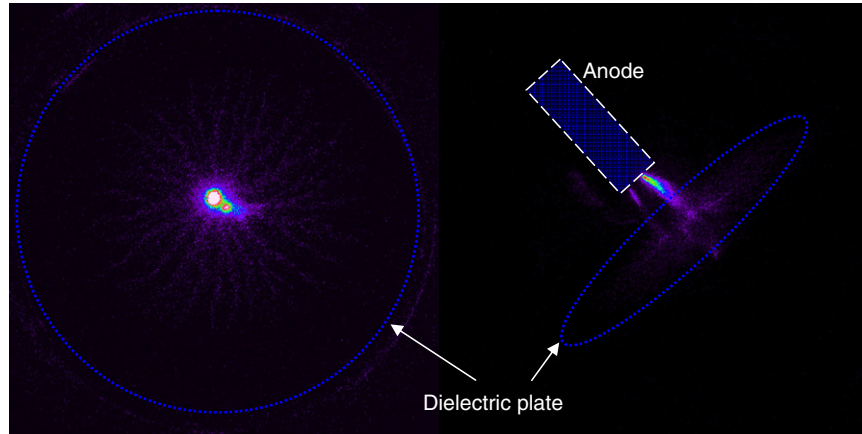


Figure 11. CCD images of the event corresponding to a first current peak when the camera is set up underneath the reactor (left image) and in perspective from an angle (right image). (Colour online.)

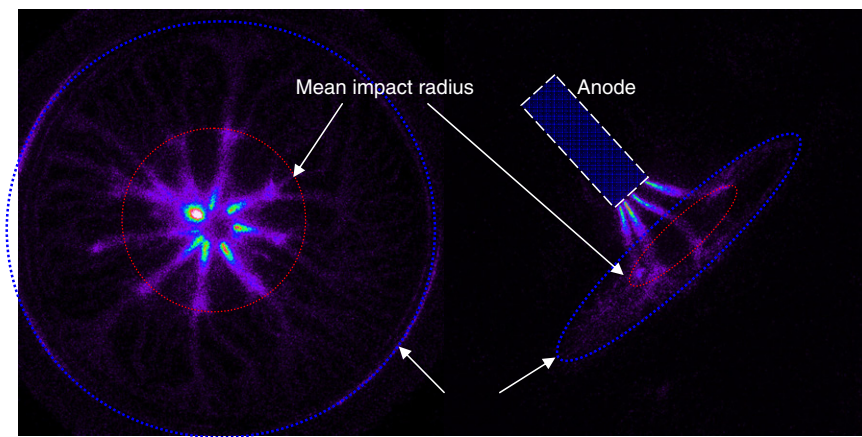


Figure 12. CCD images of the event corresponding to a second current peak when the camera is set up underneath the reactor (left image) and in perspective from an angle (right image). (Colour online.)

Mean impact radii were calculated for several current peaks and were related to the applied voltage at which the current peaks occur (V_{peak}). We realized these measurement for three applied voltage amplitudes: $V_a = 15, 18$ and 20 kV.

Increasing the applied voltage amplitude V_a , reduces the average time of the first peak appearance. This is a well-known effect in DBDs, which is due to the fact that the total amount of transferred charge in cycles of the applied voltage increases with applied voltage. This results in greater residual surface charge on the dielectric surface, causing the first peak to ignite sooner. To avoid this shift and make an overall comparison for different V_a , $\langle V_{1\text{st peak}} \rangle$, the average voltage at which the first peak occurs for one V_a , was estimated and subtracted from each V_{peak} [9].

The graph in figure 13 clearly shows an increasing linear relation of the envelope of the points. It must be noticed as well that data for the three applied voltage amplitudes 15, 18 and 20 kV match. Finally, one can see that filaments corresponding to peaks of different ranks are in different voltage ranges of V_{peak} and different length ranges of the mean impact radius. Beyond the fourth peak, we observed that the dielectric plate is not large enough for continuing the tendency shown in figure 13. Filaments which cannot reach the dielectric plate anymore may reach the side of the reactor (also made of Pyrex).

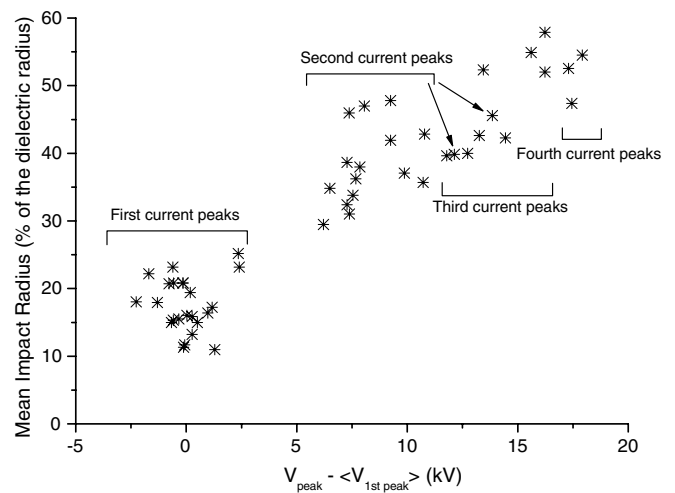


Figure 13. Mean impact radius length versus the corrected voltage.

The increase in the mean impact radius in one positive half-cycle with the rank of the current peak is due to the deposition of charges by the previous discharges in the same positive half-cycle; this is confirmed by a simple electrostatic model in section 5 and also discussed in section 6.

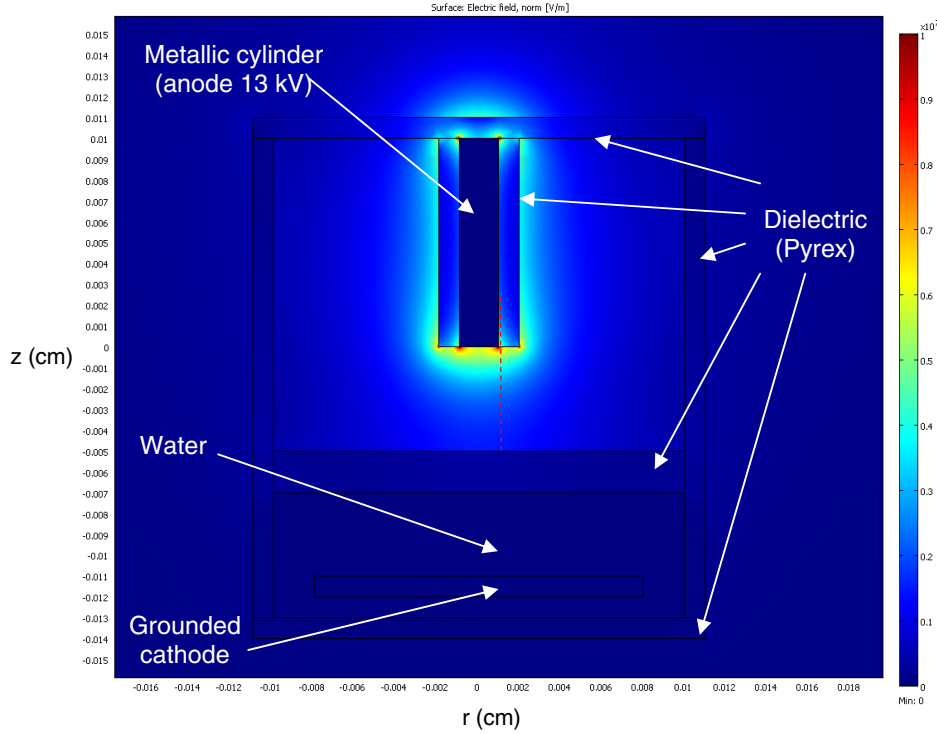


Figure 14. Electric field (V m^{-1}) given by the electrostatic model without surface charge on the dielectric plate. Red dashed line: line along which we take the norm of the electric field in figure 15.

5. Electrostatic modelling

A numerical 2D-axisymmetric electrostatic model simulation of the reactor has been performed using COMSOL Multiphysics® to validate the measured transferred charge by a microdischarge and to understand the physical impact of the charges deposited on the dielectric plate on the transient filamentary discharges (figure 14).

Breakdown occurs when the metallic electrode is at about 13 kV. As seen previously, the breakdown voltage decreases for the subsequent microdischarges due to the deposition of charges on the dielectric. Assuming a quasi-uniform distribution of the negative charge over the dielectric during the negative cycle, the potential difference between the metallic electrode and the water electrode then has to be on the order of 13 kV for a first current peak. Under this assumption, we fixed the metallic electrode at 13 kV and placed no charge upon the dielectric surface. Having defined the boundary conditions for the simulation domain, we can obtain the Laplacian electric field in the gaseous gap (figure 14), and we can also define a positive surface charge density upon the dielectric to model the screening of the electrostatic field by the transferred positive charges of a first current peak. This surface charge density is not homogeneous [3, 17] and is assumed to be Gaussian:

$$\sigma_s(r) = \frac{Q}{2\pi\sigma^2} e^{-r^2/2\sigma^2}, \quad (1)$$

where σ is the characteristic length of the charge deposit (which can be chosen according to the experimental results) and Q is the total amount of charge (integration of (1) from $r = 0$ to infinity). From the measured mean impact radius of the first

peaks from figure 13, we can take $\sigma = 2 \times 10^{-3}$ m and look for the results with different Q . By running this simulation, we found that Q must be ~ 10 – 100 nC to obtain an electric field comparable to the electric field applied. This corresponds well to the values we obtained experimentally. In figure 15 we plot the results of the simulation for the electric field along the red dashed line of figure 14 (called arc-length) from the place of the maximum Laplacian electric field (at the edge of the metallic cylinder electrode) to the dielectric plate, for different Q .

We can see in figure 15 that the value of Q for which the maximum field (at the metallic edge) is the most shielded is 40 nC. The electric field increases at this point for lower or higher Q , with the latter case corresponding to an inversion of the field. This fits well with the experimental results, as indeed $Q = 40$ nC is the average charge transferred by a first current peak.

One also can see in figure 15 that the electric field on the dielectric surface increases significantly with Q and can exceed the breakdown voltage. The physics is then no longer static and involves the triggering of surface discharges. Of course, a dynamic simulation would be necessary to completely model this behaviour.

6. Discussion

From this study, it follows that for discharges generated in air at atmospheric pressure using this particular reactor driven by a 50 Hz sine wave power supply, only a few current peaks appear in the positive half-cycle of the applied voltage (similar to the results of [4] for another DBD configuration). The electrical measurements combined with the imaging

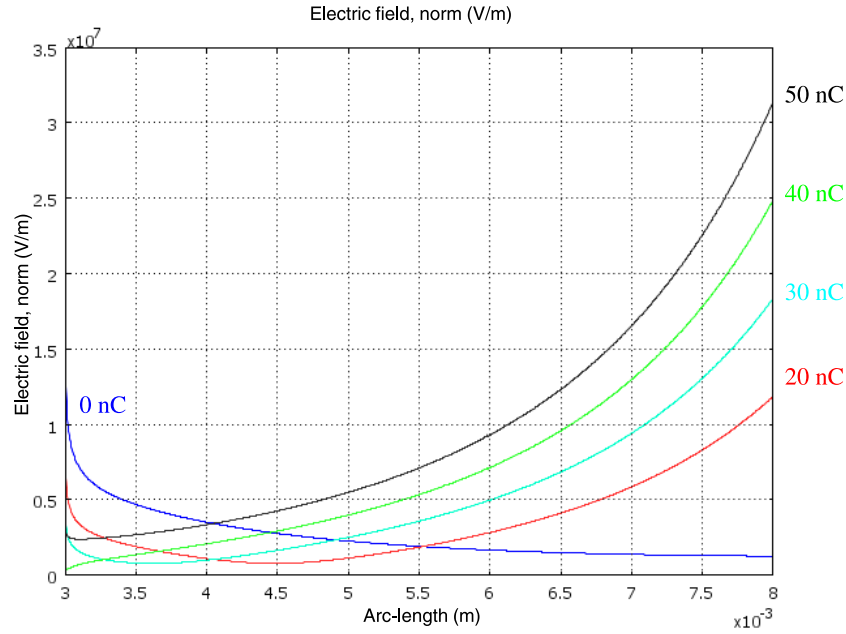


Figure 15. Norm of the electric field for several total amounts of charge upon the dielectric. Arc-length = 3 mm corresponds to the edge of the metallic electrode and Arc-length = 8 mm corresponds to the dielectric plate with a Gaussian surface charge distribution (1), with $\sigma = 2 \times 10^{-3}$ m.

diagnostics revealed that the first current peak corresponds mostly to a single centred filament in the gaseous gap (section 4, figure 11). All of the subsequent peaks always correspond to a bunch of filaments in the gaseous gap (section 4, figure 12). This reveals the generation of several transient discharges in a time window smaller than 50 ns, according to the current peak measurements (i.e. figure 4), which is a time scale effectively instantaneous compared with the characteristic rising time of the applied voltage (~ 1 ms for the results presented herein). Photon radiation coupling may explain the correlation between so many filaments over such a time scale. Guaitella *et al* [8] explained that an initial filament triggers the others by irradiating the dielectric plate, but we can also imagine that additional radiative processes by a first streamer play a role, such as photoionization in the high-field region.

Furthermore, we see in figure 12 that the subsequent discharges following a first peak do not reach the centre of the dielectric anymore. More precisely, the mean impact radius was measured as an increasing function of V_{peak} (see figure 13). This effect on the layout of the plasma filaments is due to the charges deposited on the dielectric. It must be noted that the behaviour of the transient filaments and the measurements of the charge transferred by a first current peak are in good agreement with a simple electrostatic model discussed in section 5. This model also confirms that the screening mechanism of a DBD, that is the deposition of charges on a dielectric, changes the entire field configuration in the gaseous gap as well as the propagation of subsequent streamers. As stated in [16], in a typical discharge situation there is a competition between individual microdischarges for the available surface area. In the situation we encountered in this paper, this competition is dominated by the residual charges already deposited on the dielectric by the previous

discharges, and this effect makes the filaments of successive current peaks extend farther and farther from the centre of the dielectric to deposit their charge. It has to be noted that in this study we did not observe a single sequence of current peaks in the same positive half-cycle of the applied voltage. It was impossible because the readout time of the CCD was too long.

Moreover, figure 9 shows that, for the first peak in the positive half-cycle, the maximum current increases with V_{peak} , the applied voltage at which the current peak appears. Therefore, because the charge transferred is related to the screened field, in this experimental configuration, the discharge does not occur for a constant voltage in the gaseous gap (meaning the gap above the deposits of charge). Equivalently, the discharge does not occur at a constant field, presumably the breakdown field, in the high-field region near the metal cylinder edge. Indeed, if the microdischarge occurred at a constant voltage in the gaseous gap (presumably the breakdown voltage in the considered geometry), it would always generate the same current necessary to transfer sufficient charge to screen the electric field. We propose that the ignition time of the first current peak (the current–voltage relationship of which is shown in figure 9) corresponds to the efficiency and the stochasticity of the initiation of the ‘pre-breakdown phase’ described in [3, 17] or the ‘Townsend phase’ described in [18]. In the configuration used in this paper, the pre-breakdown phase activity occurs at the edge of the metallic cylinder electrode, where the radius of curvature is smallest and hence where the electric field is strongest. The moment of the beginning of the Townsend phase is suspected here to be influenced by the charge deposited on the dielectric, as surface charge can affect the secondary emission from the dielectric. The current of this microdischarge is then determined by the

electric field at the metallic electrode when the microdischarge finally occurs.

The impact of the amount of charge deposited on the dielectric plate on (1) the multi-filament behaviour and (2) the intensity of the discharges cannot be resolved solely by using a static model; a dynamic model of streamer propagation toward a charged dielectric plate is required.

7. Conclusion

- The electrical measurements combined with the imaging diagnostics revealed that the first current peak in the positive half-cycle corresponds mostly to a single centred filament in the gaseous gap but all of the subsequent peaks always correspond to organized multi-filamentary discharges in the gaseous gap. Both kinds of discharges are strongly branching over the dielectric material.
- The spatial organization of plasma filaments is strongly affected by the deposited charges on the dielectric material. A first discharge in the positive half-cycle deposits charges close to the centre of the dielectric plate. Subsequent discharges are affected by these surface charges and plasma filaments then impact the dielectric farther and farther from the centre of the dielectric plate.
- The first current peak ignition time corresponds to the efficiency and the stochasticity of the initiation of the 'pre-breakdown phase' and consequently the amount of transferred charges in the positive half-cycle is correlated with the voltage at the metallic electrode when a first current peak occurs. This has an impact on the temporal organization of discharges in one positive half-cycle.
- A model of the streamers dynamics in such an experimental configuration is required to fully understand the transition between the volume streamers and the surface streamers and to obtain a time resolved model of the deposition of charges over the dielectric plate.

Acknowledgment

The authors would like to thank Dr David Pai (EM2C, Ecole Centrale Paris) for his help to improve the quality of this paper.

References

- [1] Kogelschatz U, Eliasson B and Egli W 1997 *J. Physique IV C* **4** 47–66
- [2] Eliasson B and Kogelschatz U 1991 *IEEE Trans. Plasma Sci.* **19** 309–23
- [3] Kozlov K V, Wagner H-E, Brandenburg R and Michel P 2001 *J. Phys. D: Appl. Phys.* **34** 3164–76
- [4] Guikema J, Miller N, Niehof J, Klein M and Walhout M 2000 *Phys. Rev. Lett.* **85** 3817
- [5] Golubovskii Yu B, Maiorov V A, Behnke J and Behnke J F 2002 *J. Phys. D: Appl. Phys.* **35** 751–61
- [6] Klein M, Miller N and Walhout M 2001 *Phys. Rev. E* **64** 026402
- [7] Chirokov A, Gutsol A, Fridman A, Sieber K D, Grace J M and Robinson K S 2004 *Plasma Sources Sci. Technol.* **13** 623–35
- [8] Guaitella O, Thevenet F, Guillard C and Rousseau A 2006 *J. Phys. D: Appl. Phys.* **39** 2964–72
- [9] Allegraud K, Guaitella O and Rousseau A 2007 *J. Phys. D: Appl. Phys.* **40** 7698–706
- [10] Eliasson B and Kogelschatz U 1991 *IEEE Trans. Plasma Sci.* **19** 1063–77
- [11] Manley T C 1943 *Trans. Electrochem. Soc.* **84** 83–96
- [12] Celestin S, Allegraud K, Canes-Boussard G, Leick N, Guaitella O and Rousseau A 2008 *IEEE Trans. Plasma Sci.* **36** 1326–7
- [13] Murooka Y, Takada T and Hidaka K 2001 *IEEE Electr. Insul. Magn.* **17** 6–16
- [14] Bertein H 1973 *J. Phys. D: Appl. Phys.* **6** 1910–9
- [15] Zhu Y, Takada T, Sakai K and Tu D 1996 *J. Phys. D: Appl. Phys.* **29** 2892–900
- [16] Kogelschatz U 2002 *IEEE Trans. Plasma Sci.* **30** 1400–8
- [17] Yurgelenas Yu V and Wagner H-E 2006 *J. Phys. D: Appl. Phys.* **39** 4031–43
- [18] Braun D, Gibalov V and Pietsch G 1992 *Plasma Sources Sci. Technol.* **1** 166–74